

ENGINEERING DRAWING

EDR 101

SOLID OBJECT

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COURSE CONTENT

Lecture	Topics
01	Introduction to Engineering Drawing
02	Lettering, Numbering, Dimensioning, Drawing Scales
03	Plane Geometry: Pentagon, Hexagon, Octagon, Ellipse, Parabola, Hyperbola
04	Solid Object: Polyhedra, Prism, Pyramid, Cone, Cylinder, Sphere, Frustum, Truncated
05	Projection: Orthographic, Isometric, Oblique
06	Drawing isometric projections of solids; Drawing isometric views of objects
MID TERM – I	
07	Drawing orthographic projections of solids and different objects
08	Introduction to AutoCAD: Interface, Basic command, Layout, Creating different types of drawing, Dimension & Annotation, Printing & Publishing
09	Course wrap-up and evaluation
FINAL EXAMINATION	

WHAT IS PROJECTION?

As per optical physics, an object is seen when the light rays, called visual rays, coming from the object strike the observer's eye.

If an imaginary transparent plane is introduced such that the object is in between the observer and the plane, the image obtained on the screen is as shown in Fig.5.1. This is called perspective view of the object. Here, straight lines (rays) are drawn from various points on the contour of the object to meet the transparent plane, thus the object is said to be projected on that plane.

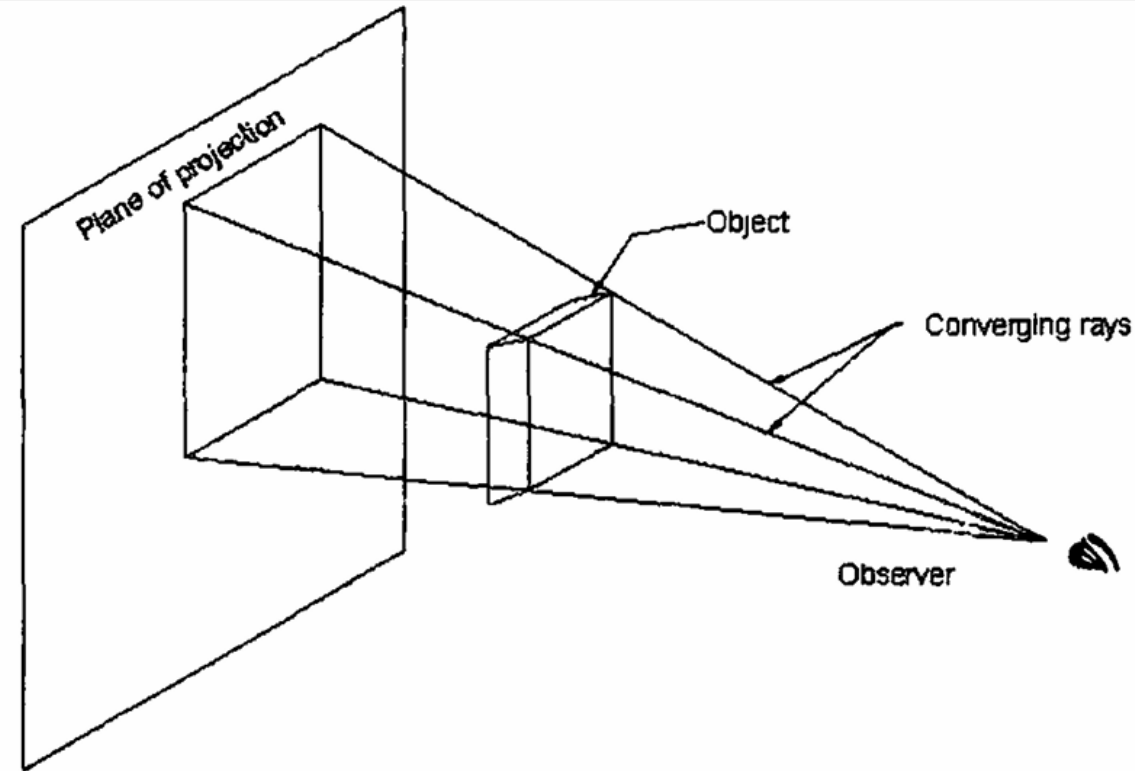


Fig. 5.1 Perspective Projection

TYPES OF PROJECTIONS

1. Pictorial projection: The Projection in which the description of the object is completely understood in one view is known as a pictorial projection. They have the advantage of conveying an immediate impression of the general shape and details of the object, but not its true dimensions or sizes.

- (i) Perspective projection
- (ii) Isometric projection
- (iii) Oblique projection

2. Orthographic Projection: 'ORTHO' means right angle and orthographic means right angled drawing. When the projectors are perpendicular to the plane on which the projection is obtained, it is known as orthographic projection.

ORTHOGRAPHIC PROJECTION

- VP (Vertical Plane) and HP (Horizontal Plane) are called the Principal planes of projection or reference planes.
 - They are always transparent and at right angles to each other.
 - The projection on VP is designated as Front view, and the projection on HP as Top view.
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- **VP (Vertical Plane):** Perpendicular to the ground, often used for representing front views or elevations.
 - **HP (Horizontal Plane):** Parallel to the ground, often associated with elevation or level surfaces.

ORTHOGRAPHIC PROJECTION

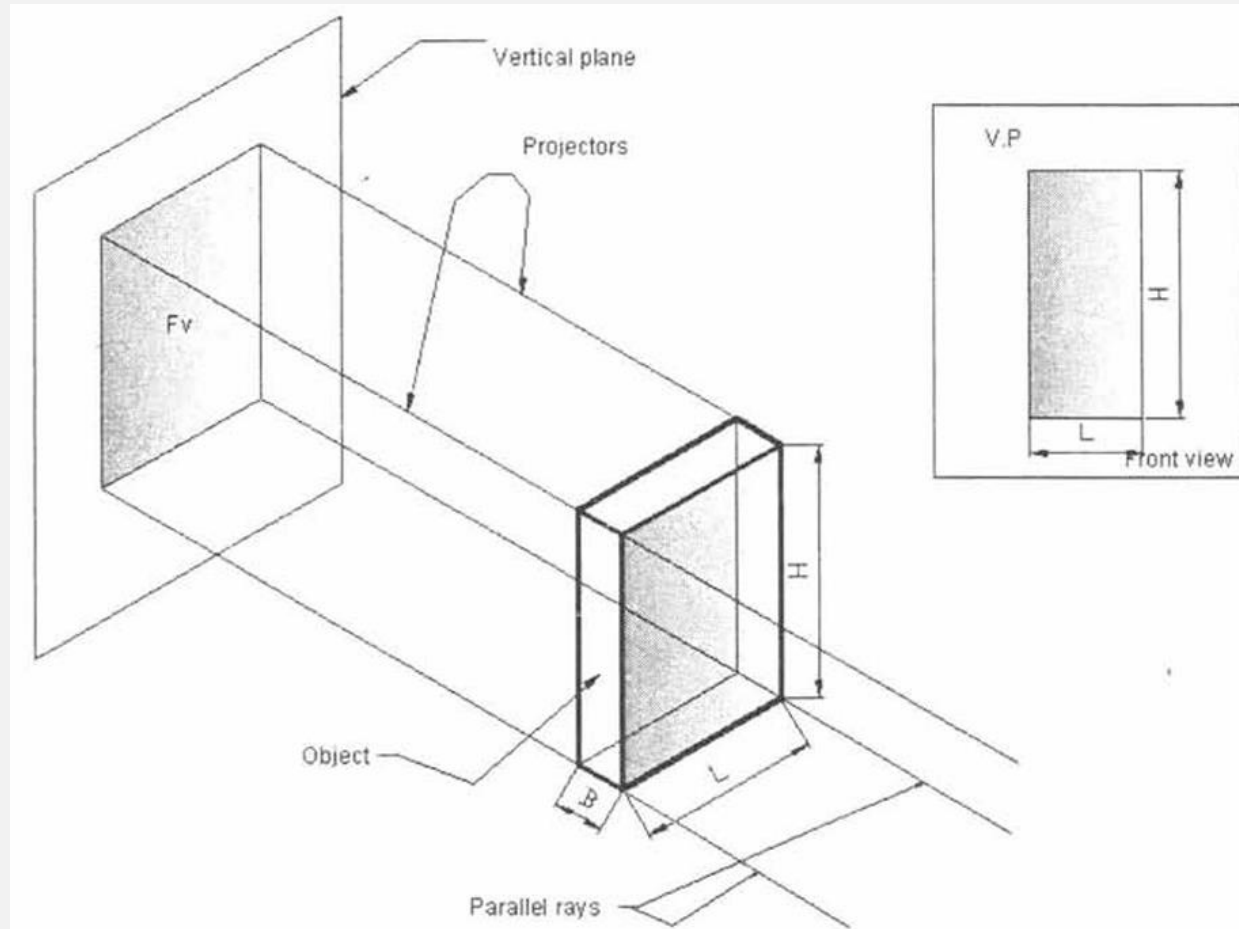


Fig. 5.2 Method of Obtaining Orthographic Front View

ORTHOGRAPHIC PROJECTION

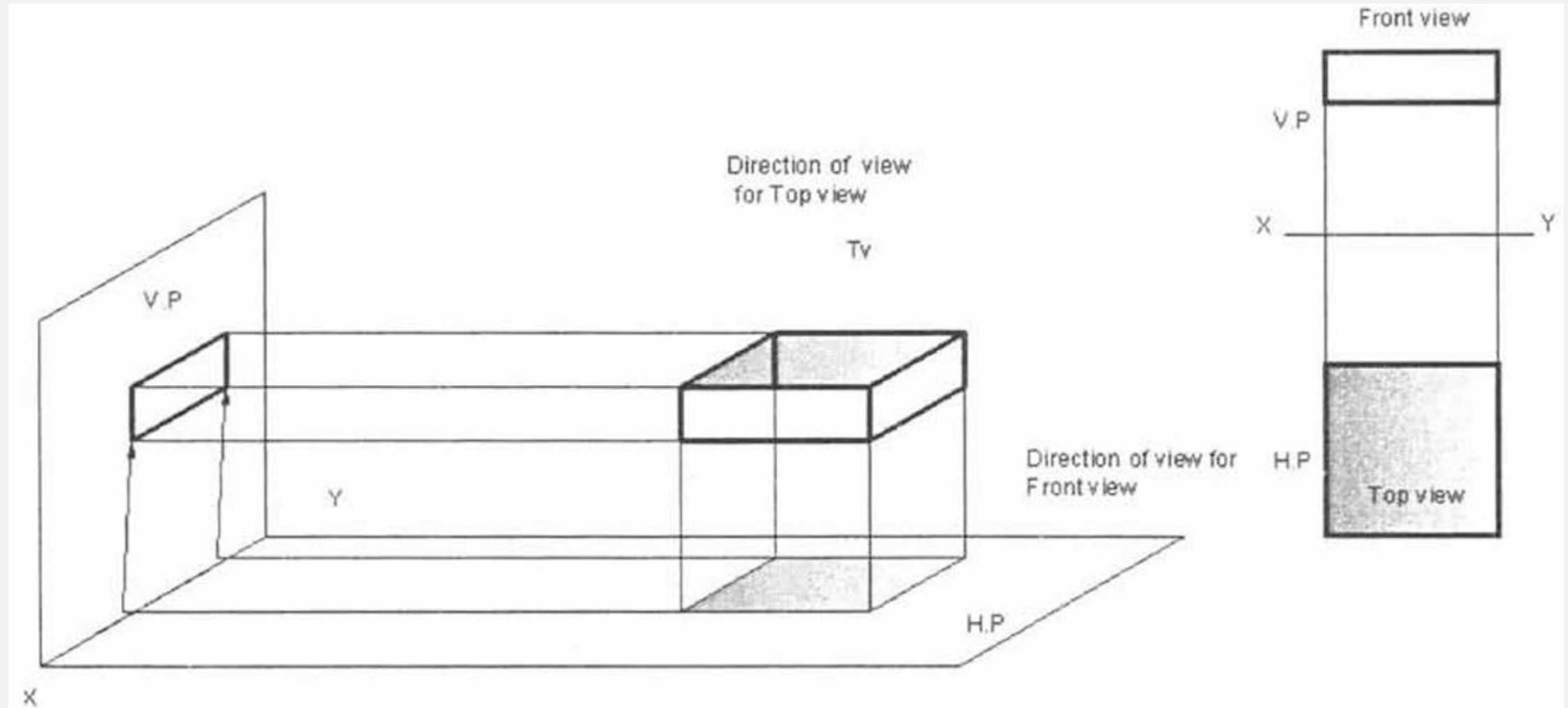


Fig. 5.3 Method of Obtaining Orthographic Top View.

ORTHOGRAPHIC PROJECTION

- Figure 5.5 shows the two principal planes HP and VP, and another **Auxiliary vertical plane (AVP)**.
- AVP is perpendicular to both VP and H.P.
- The projection on the AVP as seen from the left of the object and drawn on the right of the front view, is called the left side view.

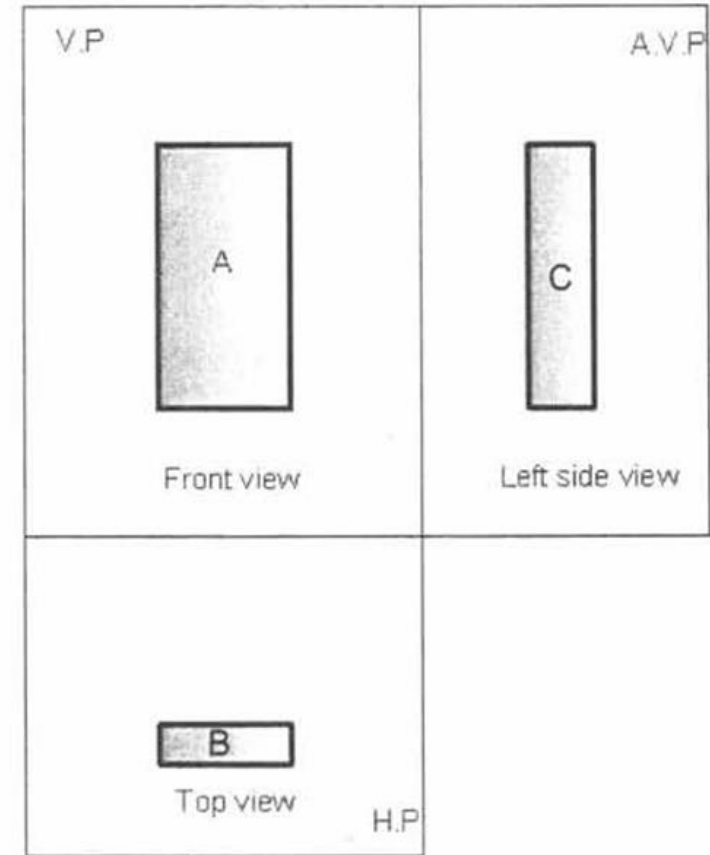
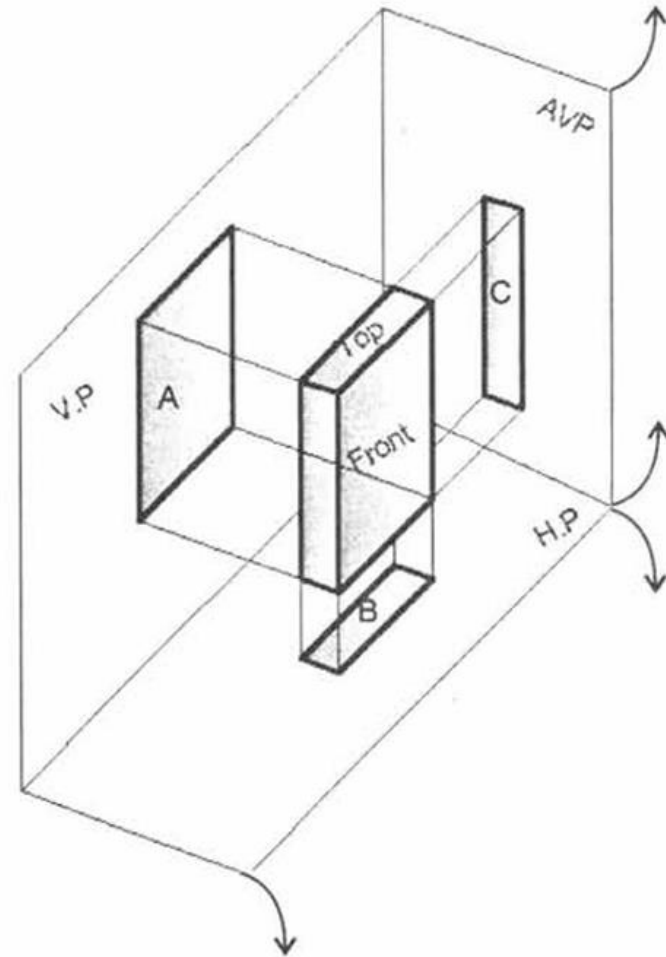
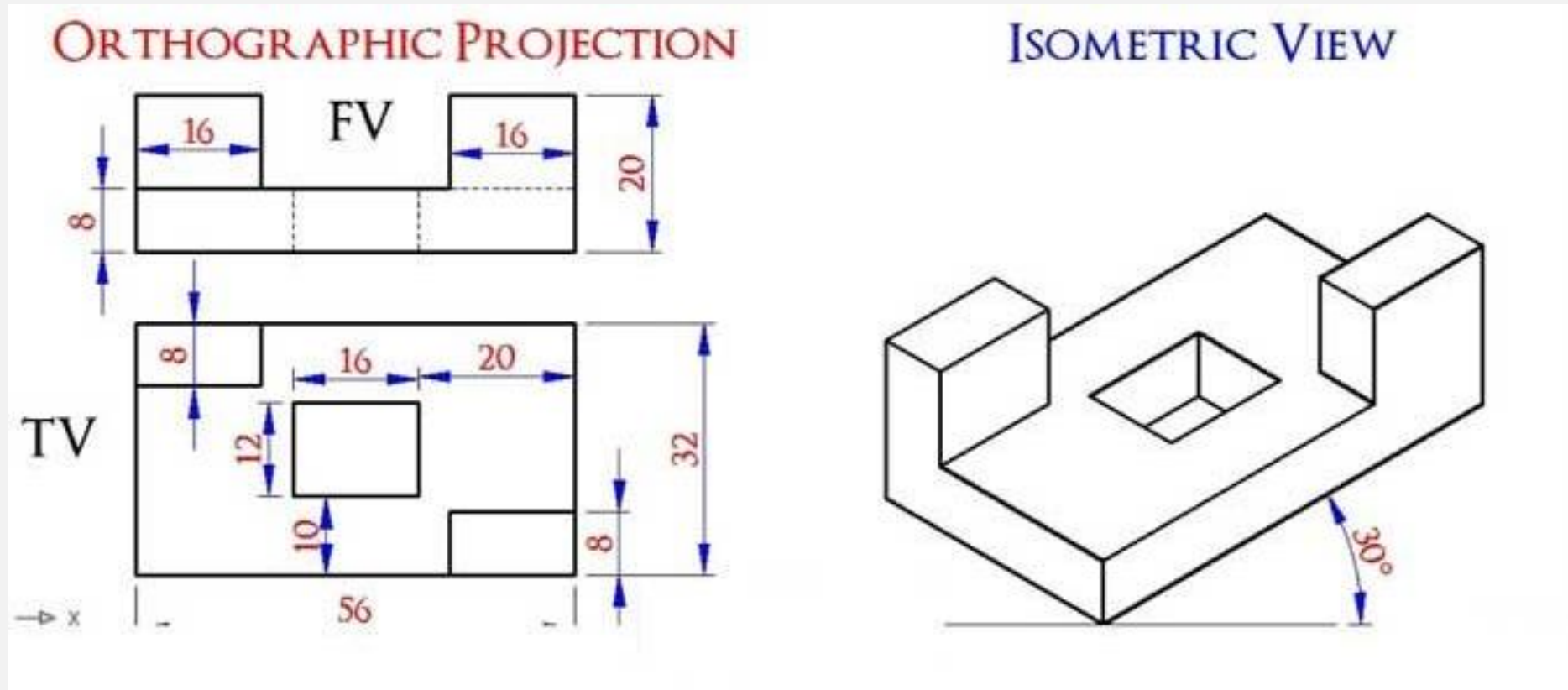


Fig. 5.5 Orthographic Projection of Front, Top and Side views

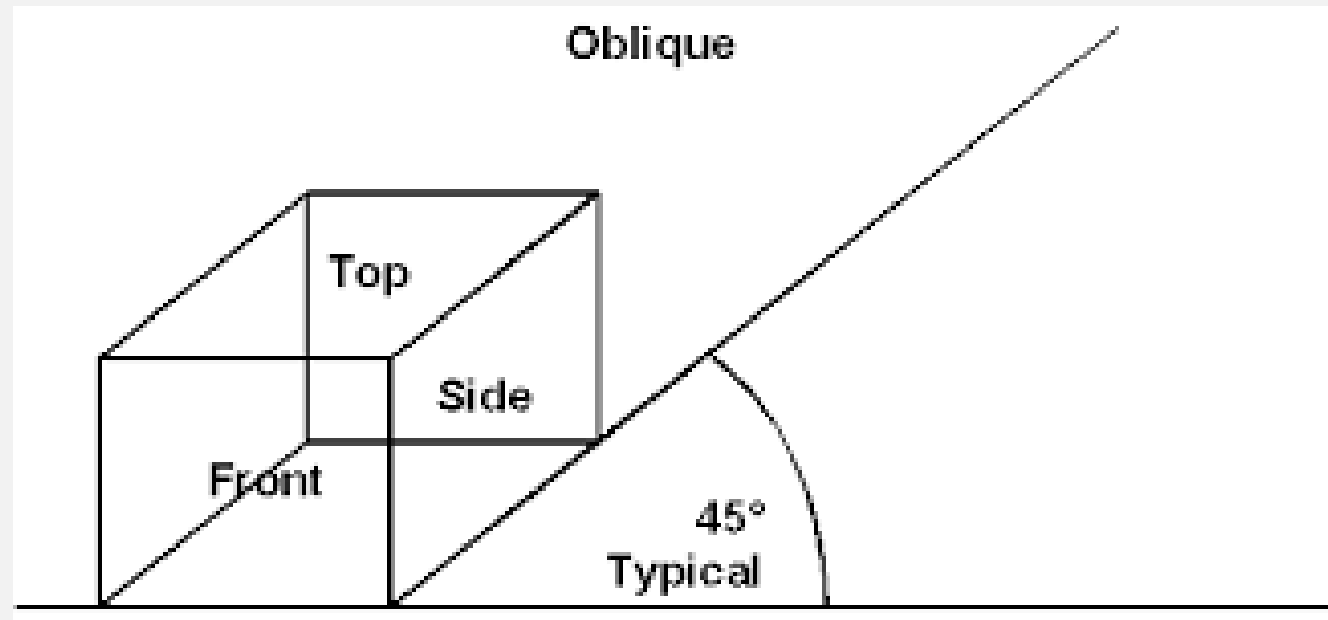
ISOMETRIC PROJECTION

- Isometric projection is drawn by establishing a vertical axis and two 30-degree angled axes to create a 3D view on a 2D surface, ensuring all lines are parallel



OBLIQUE PROJECTION

- Oblique projection is a 3D drawing method where the front face of an object is drawn in true shape and size, with depth lines projected at a 45-degree angle.
- Draw the front view, extend 45° lines from all corners, and connect them to create depth.



THANK YOU